IOP Conference Series:
Earth and
Environmental Science

PURPOSE-LED PUBLISHING™

PAPER • **OPEN ACCESS**

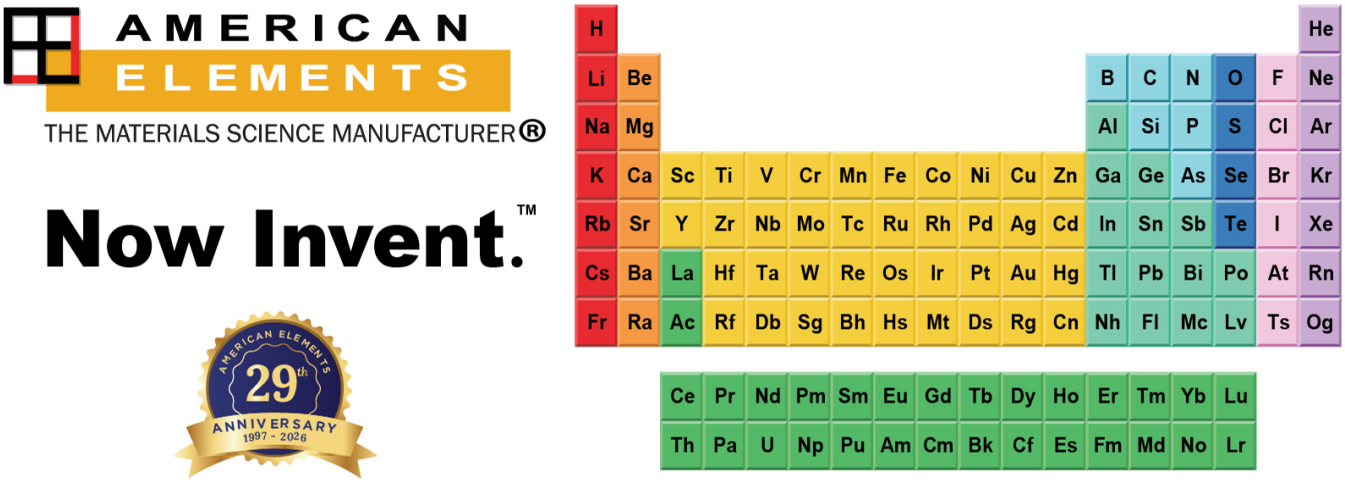
Study on the existence and characteristics of Sonokeling (*Dalbergia latifolia* Roxb) as an Appendix II CITES Wood

To cite this article: W Dwianto *et al* 2019 *IOP Conf. Ser.: Earth Environ. Sci.* **374** 012063

View the article online for updates and enhancements.

You may also like

- A bibliometric analysis of wildlife governance in relation to African elephant conservation and community participation
Tanyaradzwa Mundoga, Walter Musakwa and Nelson Chanza
- *Glomus etunicatum* improved salt tolerance in *Dalbergia latifolia* Roxb. through physiological adjustment
Delvian
- Isolation and characterization of marine bacteria from macroalgae Gracilaria salicornia and Gelidium latifolium on agarolitic activity for bioethanol production
M Kawaroe, I Pratiwi and A Sunudin



This content was downloaded from IP address 114.12.12.218 on 09/11/2026 at 09:58

IOP Conf. Series: Earth and Environmental Science **374** (2019) 012063 doi:10.1088/1755-1315/374/1/012063

Study on the existence and characteristics of Sonokeling (*Dalbergia latifolia* Roxb) as an Appendix II CITES Wood

W Dwianto[1], A Bahanawan[1], S S Kusumah[1], T Darmawan[1], Y Amin[1], D A Pramasari[1], E Lestari[1], F Akbar[1] and Sudarmanto[1]

[1]Research Center for Biomaterials, Indonesian Institutes of Sciences, Jl. Raya Bogor KM 46, Cibinong Science Center – Botanical Garden (CSC – BG) Cibinong, Bogor 16911, West Java, Indonesia

E-mail: wahyudwianto@yahoo.com

Abstract. Sonokeling (*Dalbergia latifolia* Roxb) stands are widely lost throughout the world. Sonokeling is currently classified as an Appendix II CITES (Convention on International Trade in Endangered Species of Wild Fauna and Flora) wood. Therefore, a study on the existence and characteristics of Sonokeling were expected to contribute in providing population data and basic properties of the wood. Exploration was conducted in Plot No. 25, KPH (Forest Management Unit) Gundih, Perum Perhutani Cepu, Central Java. Sonokeling stands have been planted since 1975 (more than 40 years-old) in an area of 54.20 ha, which was not productive for Teak and mixed with Mahogany. Sample was taken from upper part of stem with around 30 cm in diameter and 30 cm in length. The sample was further cutted into 2 x 2 x 2 cm for physical, 30 x 2 x 2 cm for Modulus of Elasticity (MOE) and Modulus of Rupture (MOR), and 4 x 2 x 2 cm for Compression Strength Parallel to the Grain (CSPG) tests. The results showed that specific gravity of sapwood dan heartwood were 0.79 and 0.82, respectively. The average values of MOE, MOR, and CSPG were 117,722, 1122.19, and 1022.93 kgf/cm^2, respectively. Therefore, Sonokeling can be classified into Strength Class II.

1. Introduction

Sonokeling (*Dalbergia latifolia* Roxb) is a high-quality wood in trade, known as Rosewood [1]. In Indonesia, Sonokeling is only found growing wild in the forests of Central and East Java at altitudes below 600 m above sea level, especially in rocky, infertile, and dry soils at regular intervals. It grows in groups, but not too much, in the seasonal forests that abort the leaves in the dry season [2].

Sonokeling is mainly used by the wood, which has beautiful, purple, black or purplish black patterns with reddish brown. This wood is commonly used to make furniture, cabinets, and various high-class home furnishings. Its decorative veneer is used to coat expensive plywood surfaces. Because of its good nature, Sonokeling wood is often used to make carvings and sculptures, lathe goods, musical instruments, and sports, as well as bent wood furniture such as umbrella handles, walking sticks, and others [1]. This wood is also strong and durable, so it is not in frequently used in construction such as for sills, doors, and windows, for making railroad cars, and for equipment such as hammers, plows, rakes, and grinding machines [1]. In addition to that, Sonokeling is also used to make parquet floors.

Sonokeling belongs to medium to heavy hardwoods. The specific gravity is between 0.77 – 0.86 at a moisture content of around 15%. The texture is quite smooth, with straight fiber direction and sometimes wavy. This wood is also durable, resistant to be attacked by dry wood termites and very resistant to wood rot fungi [1]. The heartwood is brown and dark purple, with very dark brown smudges

Content from this work may be used under the terms of the Creative Commons Attribution 3.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

to black. Sapwood is whitish to yellowish, 3 – 5 cm thick, clearly distinguished from heartwood [1]. Sonokeling wood is rather difficult to do by hand, but it is very easy with a machine. This wood can be shaved so that the surface is slippery, and can be peeled and sliced to make decorative veneers. This wood can also be turned, screwed, and polished with excellent results. However, this wood is difficult to be given preservatives [1].

Sonokeling is one of the most popular agroforestry plants in Indonesia. This tree is planted in an intercropping system, interspersed with various food crops. Sonokeling also becomes an agroforestry tree. Sonokeling leaves are used for animal feed and green manure [3]. Sonokeling roots are nitrogen binding and thus can improve soil fertility [4].

Sonokeling prices are comparable to Teak, even higher. Its high value has encouraged excessive harvesting so that the natural population of this tree faces extinction. Therefore, since 1998 the IUCN World Conservation Agency has included *Dalbergia latifolia* in the category Vulnerable [5,6].

Genetic resources stands of Dalbergia trees have depleted considerably all over the world. The decorative wood has been imported in Europe for several countries for furniture and interior finishing. In many areas, large Dalbergia trees become rare. In Java, for instance, it is now difficult to find an old and large Sonokeling tree. However, Sonokeling is planted on a considerable scale in agroforestry systems in India, and, on a much smaller scale, in Java [1].

Sonokeling is currently classified as an Appendix II CITES (Convention on International Trade in Endangered Species of Wild Fauna and Flora) wood, besides Agarwood (generas of *Gyrinops* and *Aquilaria*), Ramin (*Gonystylus bancanus*), and Angsana (*Pterocarpus indi*). Therefore, a study on the existence and characteristics of Sonokeling were expected to contribute in providing population data and basic properties (physical and mechanical) of the wood for classification and standardization.

2. Materials and methods

Exploration or field study was conducted in Plot No. 25, KPH (Forest Management Unit) Gundih, Perum Perhutani Cepu, Central Java on July 11, 2018, along with logging activities (figure 1). Sonokeling stands have been planted since 1975 (more than 40 years-old) in an area of 54.20 Ha, which is not productive for Teak and mixed with Mahogany.

The sample was taken from the upper part of the stem with around 30 cm in and 30 cm in length (figure 1). The sample was further cut into 2 x 2 x 2 cm for physical, 30 x 2 x 2 cm for Modulus of Elasticity (MOE) and Modulus of Rupture (MOR), and 4 x 2 x 2 cm for Compression Strength Parallel to Grain (CSPG) tests.



Figure 1. Logging activities (left) and cutting a wood sample (right).

2.1. Physical properties

Moisture content and specific gravity of wood samples were measured according to British Standard BS 373-1957 [7]. After weighing and measuring in length, width and thickness to get the initial weight (*Wo*) and volume (*Vo*) in fresh condition, the samples were dried at 103 ± 2°C for 24 hours to get oven-dry

weight (*Wod*). Moisture content was obtained by [(*Wo* – *Wod*) /*Wod*] x 100%, and specific gravity was calculated by (*Wod* / *Vo*).

2.2.Mechanical properties

Modulus of elasticity (MOE), Modulus of rupture (MOR), and Compression strength parallel to the grain (CSPG) were tested according to British Standard BS 373-1957 [7], as well. For bending tests (MOE and MOR), the wood sample was installed, and the load was given in the middle of the span. The maximum load was recorded until it was broken. For the CSPG test, the wood sample was given a compressive load parallel to the grainuntil it was damaged. MOE, MOR and CSPG values were calculated using the formula:

$$MOE\ (kgf/cm^2)=[(\Delta PL^3)/(4\ \Delta ybh^3)] \quad (1)$$

$$MOR\ (kgf/cm^2)=[(3PL)/(2bh^2)] \quad (2)$$

$$CSPG\ (kgf/cm^2)=[P/(lb)] \quad (3)$$

where P = load (kgf), ΔP= load difference (kgf), L = span distance (cm), Δy = deflection (cm), b = sample width (cm), h = sample thickness (cm), l= sample length (cm).

3. Results and discussions

3.1. Study on the existence of Sonokeling stands

Study of primary data on the existence of Sonokeling stands was conducted on July 9 – 13, 2018 through visiting several authorized and related agencies in Central Java Province, such as Ministry of Environment and Forestry, Conservation and Natural Resources Agency (BKSDA), and Perum Perhutani (Regional Division Office, Research and Development Center Cepu, and Forest Management Unit Gundih).

The results of study showed that the existence of this wood species was very difficult to find. According to Annual Report of Conservation and Natural Resources Agency [8], Sonokeling stands grow in Pagerwuning Darupodo Nature Preserve Area (33.20 ha), Wildlife Reserve Area of Mount Tunggangan (102.475 ha), Natural Tourist Park of Mount Selok (116.166 ha). Around 20 Sonokeling trees at Gunung Tunggangan have a diameter ranging between 90 and more than 300 cm. However, some of them fell due to natural disasters.

There is no export quota for nature Sonokeling wood in Central Java unless cultivation harvesting can be utilized according to its production capacity. KPH (Forest Management Unit) Gundih, BKPH Juwono, RPH Kalang got a clear cutting target for fiscal year 2018 with a planned harvesting area of 25 ha in 8 separate blocks, consisting of 2453 Sonokeling trees (1191.603 m^3), 114 Mahogany trees (73.193 m^3) and 147 Johar trees (54.458 m^3). Buyers will come to the log yard (figure 2) and use the wood for various products.



Figure 2. Log yard at KPH (Forest Management Unit) Gundih.

Recently, the selling price of Sonokeling wood is around IDR 17,000,000/m^3. For comparison, the selling price of Teak wood is IDR 6,000,000/m^3, and for Mahogany wood is IDR 4,000,000/m^3. The high selling price of Sonokeling wood is mostly due to itslimited stock, but the demand is quite high. There was no intensive cultivation effort which has an impact on Sonokeling wood supply. Therefore, serious attention is necessary to developcultivation, because Sonokeling wood is very potential and prospective.

In general, Rosewood is one of the most expensive woods. Natural stands of Dalbergia, however, need urgent protection. Loggers should take full account of the fact that the trees are usually slow growers; therefore, the cutting cycles should be very long. Sonokeling is suitable for incorporation in agroforestry systems, but to obtain straight stem, close spacing is desirable, and this means establishing monoculture plantations. Sonokeling can offer good prospects for timber production in plantations in Java, but more research is needed on silvicultural aspects.

3.2. Physical and mechanical properties

Table 1 and 2 show the moisture content and specific gravity of heartwood (table 1) and the sapwood (table 2) of Sonokeling wood samples. The results showed that the average moisture content and a specific gravity of heartwood were 49.59% and 0.82 respectively, while for the sapwood was 25.02% and 0.79 respectively. The moisture content of sapwood was lower than heartwood because the tree had been cut for a few days and left in the field before the wood sample was taken. Therefore, the moisture content values do not represent the fresh condition of the wood.

Table 1. Moisture content and a specific gravity of heartwood.

Samples	Moisture content (%)	Specific gravity
1	43.30	0.84
2	51.77	0.77
3	62.34	0.74
4	49.66	0.84
5	47.04	0.84
6	48.22	0.84
7	48.34	0.84
8	45.99	0.82
9	49.65	0.85
Average	49.59	0.82

Table 2. Moisture content and a specific gravity of sapwood.

Samples	Moisture content (%)	Specific gravity
1	21.62	0.75
2	27.64	0.76
3	22.18	0.84
4	27.57	0.75
5	26.74	0.76
6	26.17	0.75
7	24.93	0.84
8	24.39	0.84
9	23.98	0.83
Average	25.02	0.79

Table 3. Bending strength.

Samples	MOR (kgf/cm^2)	MOE (kgf/cm^2)
1	1,640.79	140,909
2	1,259.44	129,010
3	994.53	102,174
4	1,008.78	91,764
5	1,146.95	126,224
6	888.66	115,860
7	916.18	118,115
Average	1,122.19	117,722

Table 4. Compression strength parallel to grain (CSPG).

Samples	CSPG (kgf/cm^2)
1	1,147.58
2	1,042.21
3	1,166.72
4	1,170.66
5	960.51
7	1,002.49
6	1,116.71
8	872.29
9	727.20
Average	1,022.93

Table 3 and 4 show the average values of MOE, MOR, and CSPG values were 117,722, 1122.19 and 1022.93 kgf/cm^2, respectively. Therefore, Sonokeling can be classified into Strength Class II, according to table 5.

A test of Sonokeling wood from Java gave the following results for mechanical properties (density at test 790 kg/m^3, moisture content 12.5%), MOR 114 N/mm^2 or around 1,140 kgf/cm^2, MOE 270 N/mm^2 or around 2,700 kgf/cm^2, and compression parallel to grain 61 N/mm^2 or around 610 kgf/cm^2 [1]. These results meet the MOR value, but quite higher for MOE and CSPG values.

Furthermore, Soerianegara and Lemmens [1] and Martawijaya *et al*. [9] reported that the rate of shrinkage from green to oven dry is 2.9% radial and 6.4% tangential, with T/R ratio 2.2%. Sonokeling wood is durable; graveyard tests showed an average life in contact with the ground of 7.3 years long

under tropical conditions. Laboratory tests indicated that the wood was resistant to dry wood termites and very resistant to wood rot fungi. It is difficult to treat with preservatives using the vacuum-pressure method. Sonokeling wood contained 54% cellulose, 27% lignin, 10% pentosan, 1% ash and 0.6% silica. The solubility was 4.5% in alcohol-benzene, 1.8% in cold water, 5.2% in hot water, and 15.9% in 1% NaOH solution.

Table 5. Strength class of wood [10].

Strength class	Specific gravity	Absolute bending strength (kg/cm²)	Absolute compression strength (kg/cm²)
I	over 0.90	over 1100	over 650
II	0.60 to 0.90	725 to 1100	425 to 650
III	0.40 to 0.60	500 to 725	300 to 425
IV	0.30 to 0.40	360 to 500	215 to 300
V	under 0.30	under 360	under 215

4. Conclusions

The existence of this wood species was very difficult to find. There was no intensive cultivation effort which has an impact on Sonokeling wood supply. Therefore, serious attention is necessary to develop cultivation, because Sonokeling wood is very potential and prospective.

The average moisture content and a specific gravity of heartwood of the wood samples were 49.59% and 0.82 respectively, while for the sapwood was 25.02% and 0.79 respectively. The average values of MOE, MOR, and CSPG values were 117,722, 1122.19 and 1022.93 kgf/cm^2, respectively. Therefore, Sonokeling can be classified into Strength Class II.

Acknowledegments

Authors would like to thank Mr. Atang and Mr. Arif Sylvianto from Perum Perhutani KPH Gundih for their assistance and providing wood samples, as well as financial support from CITES Program FY 2018 of Research Center for Biology, Indonesian Institute of Sciences.

5. References

[1] Lemmens RHMJ and Soerianegara I (ed) 1994 *Plant Resources of South-East Asia* 5(1) Timber Trees: Major Commercial Timbers (Bogor: PROSEA)
[2] Heyne K 1987 *Useful plants Indonesia2* (in Indonesian) (Jakarta: YayasanSaranaWana Jaya)
[3] NFTA 1994 *Dalbergia latifolia: a High-valued Indian Rosewood* (Waimanalo: NFTA 94-04)
[4] Sukadi T 1994 Agroforestry-based Plantations of Dalbergia latifolia Roxb. in Java, Indonesia, (ed) Westley SB and Roshelko JM *Dalbergia* (*Proceedings of an International Workshop. Nitrogen Fixing Tree Research Reports, Special issue*)
[5] Asian Regional Workshop (Conservation & Sustainable Management of Trees, Viet Nam, August 1996) 1998 *Dalbergia latifolia* (The IUCN Red List of Threatened Species 1998)
[6] World Conservation Monitoring Centre 1998 *Dalbergia latifolia* (The IUCN Red List of Threatened Species 2006)
[7] British Standard BS 373-1957
[8] Annual Report of Conservation and Natural Resources Agency 2016 (Ministry of Environment and Forestry, Central Java)
[9] Martawijaya A, Kartasujana I, Kadir K and Prawira SA 1986 *Indonesian Wood Atlas Volume I.* (Bogor: Forest Products Research and Development Center, Forestry Research and Development Agency, Department of Forestry)
[10] Indonesian Forestry Vademecum 1976